

YOLANDA JIAN

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Education

University of Waterloo

Sept 2023 – Apr 2028

BASc. in Management Engineering (AI Option), CGPA: 3.8 / 4.0

- 2023 President's Scholarship for exceptional academic performance.
- **Relevant Coursework:** Machine Learning, Probability & Statistics, Computational Statistics & Analysis, Stochastic Models & Methods, Linear Algebra, Data Structures & Algorithms, Database & Software Design, Optimization

Work Experience

Sparkgeo

Remote

Geospatial Data Science Intern

May 2026 – Aug 2026

- Containerized pipeline stages with Docker and orchestrated a 6-stage ML pipeline on AWS Batch/SageMaker, turning 513 raw satellite scenes into 500K+ training-ready samples.
- Fine-tuned foundation-model embeddings in PyTorch and ran a 3×2 feature/classifier ablation study in scikit-learn, isolating them as the top driver of sea ice classification accuracy over raw satellite features.
- Presenting findings at the Canadian Space Agency's 2026 National Earth Observation Forum, comparing embedding-based and non-embedding baseline models against published benchmark accuracy.

Canadian Imperial Bank of Commerce (CIBC)

Toronto, ON

Software Project Coordinator Intern

Sept 2025 – Dec 2025

- Built SQL- and Excel-based forecasting models across 38 software projects to track budgets and variances, surfacing delivery risk earlier and enabling data-informed leadership decisions.
- Engineered Python-based ETL pipelines to ingest, clean, and transform project data from Jira, Confluence, and Excel, feeding Power BI dashboards used for weekly executive reporting across 50+ projects.
- Standardized and automated reporting workflows across 31 software initiatives, improving data reliability and cutting manual reporting effort by 30%.

Data Engineer Intern

Jan 2025 – Apr 2025

- Built a real-time Tableau dashboard with automated ETL workflows from Azure DevOps to track productivity metrics, reducing reporting lag by 60 minutes per day.
- Developed 5 end-to-end automation pipelines in Python, Alteryx, and Excel-based scripting to streamline data processing and reporting, cutting manual operations by 300+ hours.

Canada Revenue Agency

Toronto, ON

Data Analyst Intern

May 2024 – Aug 2024

- Wrote SQL queries using SAS and Aginity to process and analyze demographic data from 30M+ taxpayers, optimizing demographic clustering and increasing digital transformation conversion by 21%.
- Built Power BI dashboards to analyze client-platform engagement, informing a nationwide digital transition plan and reducing paper usage by 9%.
- Modeled an anomaly-detection classifier for biometric authentication risk scoring, raising flagged-case precision 30%.

Projects

Scan2Craft 3D Scan Data Pipeline | *Python, NumPy, Trimesh, scikit-image, litemapy*

- Built an end-to-end data pipeline in Python to convert 3D mesh files into structured .litematic outputs for Minecraft.
- Optimized colour-mapping logic using NumPy vectorization, reducing processing time by 40% while preserving colour accuracy across block palettes.

LLM-Powered Financial Q&A Chatbot | *Python, LangChain, HuggingFace Transformers, Streamlit*

- Developed a full-stack AI-powered question-answering system using LangChain and HuggingFace models to deliver investment insights from scraped financial knowledge bases, achieving 92% answer relevance across 22 test queries.
- Integrated FAISS vector search for document retrieval and Streamlit for an interactive UI, demonstrating practical LLM deployment and embedding pipeline design.

Technical Skills

Languages & Scripting: Python, SQL, R, Java, JavaScript, HTML/CSS, VBA

Frameworks/Libraries: PyTorch, Pandas, NumPy, scikit-learn, Matplotlib, Node.js, React

Platforms/Tools: AWS (Batch, SageMaker), Docker, Azure, Azure DevOps, Git, PostgreSQL, MySQL, Tableau, Power BI, Alteryx, Power Automate, Power Apps